

Software Architecture Diagrams Guide

Document Link:

https://docs.google.com/document/d/1GGdhAldTB4r_0-OlsMCN-qA_nrjMtQtm2pOskwBtAuU/edit?usp=sharing

A practical engineering reference for creating architecture diagrams:

- **System Context Diagram (C4 L1)**
 - External actors and system boundaries
- **Container Diagram (C4 L2)**
 - 12 containers with technology stacks and communication patterns
- **Component Diagram (C4 L3)**
 - 10 API server components with interaction patterns
- **Database ERD**
 - 14 entities with full attribute lists and relationship cardinality
- **User Flow Diagram**
 - Logistics Manager, Dispatcher, and Driver mobile app flows
- **Business Process Flow**
 - Standard dispatch process with 4 exception sub-processes
- **Sequence Diagram**
 - Dispatch assignment and GPS tracking sequences with full message flows
- **Deployment Diagram**
 - 5-layer production architecture with Kubernetes, multi-AZ, and auto-scaling
- **Data Flow Diagram**
 - GPS pipeline (real-time, persistence, analytics, alerting, audit paths)
- **Event Architecture**
 - 10 Kafka topics with producer/consumer mappings and processing patterns
- **Network Architecture**
 - DMZ, application, data, and management subnets with traffic flows
- **Security Architecture**

- Authentication flow, RBAC, multi-tenant isolation, data protection, and SIEM monitoring

- **Recommended Diagram Order**

- Phased 12-step creation guide aligned with project milestones